Prime support and growth of Stirling repair factors

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Abstract

For fixed $k \geq 1$, let F_k be the least positive integer such that $(F_k S(n+k-1,k))_{n\geq 1}$ counts the periodic points of a map, where $S(N,k)$ is a Stirling number of the second kind. We prove

$$\operatorname{rad}((k-1)!) \mid F_k \mid \operatorname{lcm}(1,\ldots,k-1).$$

Thus the prime divisors of F_k are exactly the primes below k, and the previous factorial bound can be replaced by a least common multiple. We also prove $v_p(F_k) = 1$ whenever $p < k \leq p^2$, and $v_p(F_{p^j+1}) = j$ for every prime p and $j \geq 1$. A formula for prime-power values in terms of the base-p digits of $k-1$ also yields $\log(F_k/\operatorname{rad}((k-1)!)) \sim \sqrt{k}$. These results establish Conjectures 1, 2, 3, and 4(d) of Miska and Ward. The upper bound follows from a Frobenius congruence for an integral recurrence; the prime-support obstruction comes from the repeated poles of its generating function modulo p.

1 Introduction

An integer sequence $(a_n)_{n\geq 1}$ satisfies the *Dold condition* if

$$n \mid \sum_{d\mid n} \mu(n/d) a_d \qquad (n \geq 1). \tag{1}$$

For a nonnegative integer sequence, realizability as the numbers of fixed points of the iterates of a map is equivalent to (1) together with nonnegativity of the sums in (1). Indeed, their quotients by n are precisely the required numbers of orbits of length n. A disjoint union of those finite cycles supplies a realizing map.

Miska and Ward introduced the least positive multiplier $\operatorname{Fail}(a)$ making a sequence realizable and studied it for Stirling sequences [3]. Throughout this paper,

$$a_n^{(k)} = S(n+k-1,k), \qquad F_k = \operatorname{Fail}\big((a_n^{(k)})_{n\geq 1}\big).$$

They proved that F_k is finite and divides $(k-1)!$ [3, Theorem 7]. Their Conjectures 1 and 2 predict, respectively, that every prime below k divides F_k, and that primes in $[\sqrt{k}, k)$ have valuation exactly one. Their Conjecture 4(d) predicts $v_p(F_{p^j+1}) = j$. Their Conjecture 3 predicts that $F_k/\operatorname{rad}((k-1)!)$ tends to infinity. We prove all four assertions, improve the factorial upper bound, and determine the logarithmic growth of that ratio.

Theorem 1. *For every integer $k \geq 1$,*

$$\operatorname{rad}((k-1)!) \mid F_k \mid L_{k-1}, \qquad L_{k-1} = \operatorname{lcm}(1,\ldots,k-1), \tag{2}$$

where $L_0 = 1$. More precisely, for every prime $p < k$,

$$1 \le v_p(F_k) \le \lfloor \log_p(k-1) \rfloor.$$

In addition:

1. *if $p < k \le p^2$, then $v_p(F_k) = 1$;*
2. *if $k = p^j + 1$ with $j \ge 1$, then $v_p(F_k) = j$.*

Theorem 2. *Let $\vartheta(x) = \sum_{p \le x} \log p$, where the sum is over primes. As $k \to \infty$, with $K = k - 1$,*

$$\log \frac{F_k}{\operatorname{rad}(K!)} = \vartheta(\sqrt{K}) + O\big(K^{1/3}(\log K)^2\big). \tag{3}$$

Consequently,

$$\log \frac{F_k}{\operatorname{rad}((k-1)!)} \sim \sqrt{k}. \tag{4}$$

The estimate (3) is elementary. Only the passage to (4) uses the prime number theorem.

The upper bound is sharp at the prime p whenever $k = p^j + 1$. The argument for prime support applies to the entire sequence and does not require locating an individual prime-power witness. In particular, our result does not establish the proposed finite cutoff for computing F_k described in [1, Section 4.5].

2 Local congruences and the integral recurrence

For a sequence a write $D_n(a) = \sum_{d|n} \mu(n/d) a_d$. The following standard local form of the Dold condition will be useful; we include the proof to fix the indexing.

Lemma 3. *An integer sequence a satisfies the Dold condition if and only if*

$$a_{mp^r} \equiv a_{mp^{r-1}} \pmod{p^r} \tag{5}$$

for every prime p, every $r \ge 1$, and every $m \ge 1$ with $p \nmid m$.

Proof. Möbius inversion gives $a_n = \sum_{d|n} D_d(a)$. Thus

$$a_{mp^r} - a_{mp^{r-1}} = \sum_{d|m} D_{dp^r}(a),$$

which proves one implication. Conversely,

$$D_{mp^r}(a) = \sum_{d|m} \mu(m/d)\big(a_{dp^r} - a_{dp^{r-1}}\big).$$

The congruences imply $p^r \mid D_{mp^r}(a)$. Applying this at each prime factor of an arbitrary index gives (1). ∎

Lemma 4. *For every $k \ge 1$, the sequence $a^{(k)}$ satisfies $D_n(a^{(k)}) \ge 0$ for all $n \ge 1$. Consequently, F_k is its least positive Dold multiplier.*

Proof. For $k = 1$ the sequence is constant equal to one, so the assertion is immediate. For $k \geq 2$, the Stirling recurrence gives

$$a_{n+1}^{(k)} = k a_n^{(k)} + S(n+k-1, k-1) \geq 2a_n^{(k)}.$$

As $a_1^{(k)} = 1$, for $n \geq 2$ we have

$$\sum_{d=1}^{n-1} a_d^{(k)} \leq a_n^{(k)} \sum_{d=1}^{n-1} 2^{d-n} < a_n^{(k)}.$$

Therefore $D_n(a^{(k)}) \geq a_n^{(k)} - \sum_{d<n} a_d^{(k)} > 0$. Positive multiplication preserves this sign condition. □

Fix $k \geq 2$, and suppress the superscript on a_n. Define

$$P_k(z) = \prod_{j=1}^{k} (1 - jz), \qquad G_k(X) = \prod_{j=1}^{k} (X - j).$$

The usual generating identity for Stirling numbers is

$$H_k(z) := \sum_{n \geq 1} a_n z^{n-1} = \frac{1}{P_k(z)}. \tag{6}$$

It follows directly, for example, by applying the Stirling recurrence to $\sum_{t \geq 0} S(k+t, k) z^t$: this series is the corresponding series for $k-1$, divided by $1 - kz$, and the initial series for $k = 1$ is $(1-z)^{-1}$.

Lemma 5. *Set $a_0 = S(k-1, k) = 0$. There are an integer $k \times k$ matrix T, a vector $v \in \mathbb{Z}^k$, and an integer linear functional λ such that*

$$G_k(T) = 0, \qquad a_n = \lambda(T^n v) \quad (n \geq 0).$$

Proof. Write $G_k(X) = X^k + c_1 X^{k-1} + \cdots + c_k$. Multiplying (6) by $P_k(z) = 1 + c_1 z + \cdots + c_k z^k$ shows

$$a_{n+k} + c_1 a_{n+k-1} + \cdots + c_k a_n = 0 \qquad (n \geq 0).$$

For $n = 0$ this uses the coefficient of z^{k-1}, which is zero since $k \geq 2$, and the fact that $a_0 = 0$. For $n \geq 1$ it uses the coefficient of z^{n+k-1}. Take the companion transition matrix for the states $(a_n, a_{n+1}, \ldots, a_{n+k-1})^\mathsf{T}$, take the initial state as v, and let λ extract its first coordinate. The companion identity $G_k(T) = 0$ gives the result. □

The integral initialization at zero is useful: the matrix powers represent a_n itself, without a shift in the exponent.

3 The least-common-multiple upper bound

Lemma 6. *Let X, Y be commuting integer matrices. If $X \equiv Y \pmod{p^a}$, where p is prime and $a \geq 1$, then*

$$X^{p^t} \equiv Y^{p^t} \pmod{p^{a+t}} \qquad (t \geq 0).$$

Proof. Write $X = Y + p^a C$; then C commutes with Y. In the binomial expansion of $X^p - Y^p$, the terms of degree i in C, for $1 \leq i < p$, are divisible by p^{ai+1}, hence by p^{a+1}. The last term is divisible by p^{ap}, and $ap \geq a + 1$. Iteration proves the assertion, including $p = 2$. □

Lemma 7. *Fix a prime p and put*

$$m = \lceil k/p \rceil, \qquad h = \lceil \log_p m \rceil.$$

For the matrix T of Lemma 5,

$$T^{p^{h+1}} \equiv T^{p^h} \pmod{p}.$$

Proof. Every residue class modulo p occurs among $1, \ldots, k$ at most m times. Over $\mathbb{F}_p$ this implies $G_k(X) \mid (X^p - X)^m$, so $(T^p - T)^m = 0$ modulo p. Since $p^h \geq m$, raising $T^p - T$ to the power p^h gives zero. The characteristic-p binomial identity for commuting matrices gives $T^{p^{h+1}} - T^{p^h} = 0$ modulo p. ☐

Proof of the upper bound in Theorem 1. The case $k = 1$ is immediate. For $k \geq 2$, fix a prime p and use the notation of Lemma 7. For $r \geq h + 1$, Lemma 6 gives

$$T^{p^r} \equiv T^{p^{r-1}} \pmod{p^{r-h}}.$$

Taking arbitrary positive powers preserves the congruence. Applying $\lambda(\,\cdot\, v)$ therefore gives

$$a_{tp^r} \equiv a_{tp^{r-1}} \pmod{p^{r-h}} \qquad (t \geq 1,\ r \geq h+1).$$

Multiplication by p^h supplies the required local modulus p^r. For $1 \leq r \leq h$ the same conclusion is automatic after multiplication by p^h.

If $p \geq k$, then $m = 1$ and $h = 0$. If $p < k$, then

$$h = \lfloor \log_p (k-1) \rfloor. \tag{7}$$

Indeed, for the integer on the right, $p^h < k \leq p^{h+1}$, and hence $p^{h-1} < \lceil k/p \rceil \leq p^h$. The product of these local multipliers is

$$\prod_{p<k} p^{\lfloor \log_p (k-1) \rfloor} = L_{k-1}.$$

Lemmas 3 and 4 show that this is a realizability multiplier. The least multiplier divides every multiplier: it is the least common multiple of the denominators of the rational numbers $D_n(a)/n$ in lowest terms. Consequently $F_k \mid L_{k-1}$. ☐

4 Every prime below k is necessary

Proof of the lower bound in Theorem 1. Let $p < k$, and suppose that $p \nmid F_k$. The local congruences for $F_k a$, reduced modulo p, imply

$$a_{pn} = a_n \quad \text{in } \mathbb{F}_p \qquad (n \geq 1). \tag{8}$$

Here one writes $n = tp^r$ with $p \nmid t$ and applies the local congruence at level $r + 1$, cancelling the nonzero scalar F_k modulo p. Iteration gives $a_{p^h n} = a_n$ for every $h \geq 0$.

Choose h as in Lemma 7 and set $B = T^{p^h}$ modulo p. Then $B^p = B$, whence $B^{n+p-1} = B^n$ for every $n \geq 1$. The matrix representation and (8) imply

$$a_{n+p-1} = a_n \quad \text{in } \mathbb{F}_p \qquad (n \geq 1).$$

Thus the generating series $H_k(z)$ modulo p has period $p-1$, and $(1-z^{p-1})H_k(z)$ is a polynomial. By (6), this forces

$$P_k(z) \mid 1 - z^{p-1} \quad \text{in } \mathbb{F}_p[z].$$

There is no numerator cancellation because the numerator in (6) is the constant one. But $k > p$ means that the factors with $j = 1$ and $j = p+1$ both occur, so $(1-z)^2 \mid P_k(z)$ modulo p. The polynomial $1 - z^{p-1}$ is squarefree: its derivative in characteristic p is z^{p-2}, which has no common root with it. This is a contradiction, also when $p = 2$. Hence $p \mid F_k$ for every $p < k$. $\square$

5 Prime-power values and exact valuations

The next formula extends the first-level congruence $S(p+k-1,k) \equiv \lceil k/p \rceil \pmod{p}$ of Miska and Ward [3, Lemma 10].

Lemma 8. *Let p be prime, $k \geq 1$, and $r \geq 0$. Write*

$$k - 1 = \sum_{i \geq 0} d_i p^i, \qquad 0 \leq d_i < p.$$

Then

$$S(p^r + k - 1, k) \equiv \prod_{i=1}^{r} (d_i + 1) \pmod{p}, \tag{9}$$

where the empty product is one.

Proof. Write $k = qp + c$, with $0 \leq c < p$, and put $R = (p^r - 1)/(p-1)$. Over $\mathbb{F}_p$ we have

$$P_k(z) = (1 - z^{p-1})^q P_c(z), \qquad P_0(z) = 1.$$

If $1 \leq c < p$, every coefficient of $1/P_c(z)$ at a nonnegative degree e divisible by $p-1$ equals one modulo p. Indeed, the coefficient is $S(e+c,c)$, and inclusion–exclusion gives

$$S(e+c,c) = \frac{1}{c!} \sum_{j=1}^{c} (-1)^{c-j} \binom{c}{j} j^{e+c} \equiv S(c,c) = 1 \pmod{p}.$$

Here $c!$ is invertible modulo p and Fermat's congruence applies to each $j \in \{1, \ldots, c\}$. Thus, when $c > 0$ and $q \geq 1$, the coefficient of $z^{(p-1)R}$ in $1/P_k(z)$ is

$$\sum_{t=0}^{R} \binom{q+t-1}{t} = \binom{q+R}{R} \pmod{p}.$$

When $c > 0$ and $q = 0$, this coefficient is one, which agrees with the final binomial expression. If $c = 0$, only the degree-zero coefficient of $1/P_0(z)$ contributes, and the answer is $\binom{q+R-1}{R}$. Putting $J = \lceil k/p \rceil - 1 = \lfloor (k-1)/p \rfloor$ combines both cases into

$$a_{p^r}^{(k)} \equiv \binom{J+R}{R} \pmod{p}. \tag{10}$$

The base-p digits of J are $d_1, d_2, \ldots$, and those of R are one in positions $0, \ldots, r-1$ and zero thereafter. If all $d_1, \ldots, d_r$ are at most $p-2$, adding J and R produces no carry. Lucas's coefficient formula therefore evaluates (10) as $\prod_{i=1}^{r}(d_i + 1)$. If one of these digits is $p-1$, take the first such digit. There was no earlier carry, and the corresponding digit of $J + R$ is zero. Its Lucas factor is $\binom{0}{1} = 0$, and the proposed product is zero as well. For completeness, the coefficient formula used here follows by comparing coefficients in

$$(1+x)^N = \prod_{i\geq 0}(1+x^{p^i})^{N_i} \quad \text{in } \mathbb{F}_p[x],$$

where N_i are the base-p digits of N. $\square$

Corollary 9. *Let $p < k$, put $h = \lfloor \log_p(k-1) \rfloor$, and use the digits in Lemma 8. If $1 \leq r \leq h$, $d_r \neq 0$, and $d_i \neq p-1$ for every $1 \leq i < r$, then $v_p(F_k) \geq r$. In particular, if $d_i \neq p-1$ for every $1 \leq i < h$, then*

$$v_p(F_k) = h.$$

Proof. Subtracting successive formulas in (9) gives

$$a^{(k)}_{p^r} - a^{(k)}_{p^{r-1}} \equiv d_r \prod_{i=1}^{r-1}(d_i + 1) \pmod p.$$

Under the stated conditions this is nonzero. The local Dold congruence at level r forces $p^r \mid F_k$. At $r = h$, the highest digit d_h is nonzero, and the upper bound in Theorem 1 gives equality. $\square$

Remark 10. Lemma 8 also supplies an explicit witness for full prime support. For $p < k$, let $s \geq 1$ be the first position with $d_s \neq 0$. Then $a^{(k)}_{p^s} - a^{(k)}_{p^{s-1}} \not\equiv 0 \pmod p$, with $p^s < k$. This proves the prime-support assertion by a second method and locates one necessary prime-power factor below k. It does not locate the maximum possible valuation of the repair factor.

Proof of Theorem 1(1) and (2). When $p < k \leq p^2$, we have $h = 1$, so the digit restrictions in Corollary 9 are empty and $v_p(F_k) = 1$. When $k = p^j + 1$, the digits of $k - 1 = p^j$ are zero below position j and one at position j. The same corollary gives $v_p(F_k) = j$. In particular, for $p = 2$ the corollary also gives $v_2(F_k) = h$ at both $k = 2^h + 1$ and $k = 2^h + 2$, for every $h \geq 1$. $\square$

6 Growth beyond the squarefree factor

Proof of Theorem 2. Put $K = k - 1$ and $E_k = \log(F_k/\operatorname{rad}(K!))$. By Theorem 1,

$$\begin{aligned} 0 \leq E_k &\leq \sum_{j=2}^{\lfloor \log_2 K \rfloor} \vartheta(K^{1/j}) \\ &= \vartheta(\sqrt{K}) + O\big(K^{1/3}(\log K)^2\big). \end{aligned}$$

For the error estimate, there are $O(\log K)$ terms with $j \geq 3$, and each is at most $K^{1/3} \log K$, by the elementary inequality $\vartheta(x) \leq x \log x$.

For the lower bound, consider primes

$$K^{1/3} < p \leq \sqrt{K}.$$

For each such prime, $h = \lfloor \log_p K \rfloor = 2$. Corollary 9 gives $v_p(F_k) = 2$ unless the digit d_1 in $K = d_2 p^2 + d_1 p + d_0$ is $p - 1$. In that exceptional case, $t = d_2 + 1$ satisfies

$$tp^2 - p \leq K < tp^2, \qquad 2 \leq t \leq K^{1/3} + 1. \tag{11}$$

For a fixed $t \geq 2$, at most one positive integer p can satisfy (11). Indeed, the lower endpoint for $p+1$ exceeds the upper endpoint for p, since

$$t(p+1)^2 - (p+1) - tp^2 = 2tp + t - p - 1 > 0.$$

The intervals increase with p, so they are pairwise disjoint. Consequently, at most $\lfloor K^{1/3} \rfloor + 1$ primes in the specified range are exceptional.

Every nonexceptional prime contributes at least $\log p$ to E_k. It follows that

$$\begin{aligned} E_k &\geq \vartheta(\sqrt{K}) - \vartheta(K^{1/3}) - (\lfloor K^{1/3} \rfloor + 1) \log \sqrt{K} \\ &= \vartheta(\sqrt{K}) - O(K^{1/3} \log K). \end{aligned}$$

This proves (3). The classical prime number theorem $\vartheta(x) \sim x$ [2], together with $K^{1/3}(\log K)^2 = o(\sqrt{K})$, proves (4). In particular, $F_k / \operatorname{rad}((k-1)!) \to \infty$. $\square$

7 Remaining questions

The exact value of F_k for general k is still not determined by these arguments. In particular, the finite-cutoff assertion discussed in [1, Section 4.5] would compute F_k from the denominators of $D_n(a^{(k)})/n$ only for indices n at most the largest prime power strictly below k. Our upper bound controls all prime-power valuations, but does not show that their maxima occur within that finite range.

The results above leave open parts (a)–(c) of Conjecture 4 in [3]. In particular, the assertion $v_p(F_{p^j}) = 1$ for $j > 1$ requires a sharper upper bound than the one proved here. Distinguishing those special cancellations from the general Frobenius bound appears necessary for an exact formula.

References

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